CS 4376-01 Final Report

Project: ReDuce, ReUse, ReCycle

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Figma Url:

https://www.figma.com/design/8Qpw8ucuYQcxbmHFNcTPkR/Mid-Fidelity-WireFlow?node-id=0-1&t=VlarusTiX1YlbsKt-1

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Date: 11/30/2025

A. ReDuce, ReUse, Recycle (RRR) Overview:

Our project is a redesigned and fully structured marketplace designed for Vanderbilt University students with the goal of solving the limitations of the current ReUse Vandy GroupMe chat. We also aim to provide Vanderbilt students with a platform where each student can buy and sell used items by other Vanderbilt students conveniently.

Currently, Vanderbilt students frequently buy and sell mainly through informal platforms on GroupMe and WeChat. However, these platforms are unstructured, cluttered and insecure making it hard for students to undergo their transactions easily. With these platforms, messages are easily buried and there is no reliable way for students to filter items they are interested in. In addition, users often waste time scrolling or inquiring about items that are no longer available. Also, the absence of authentication exposes students to scams, bots and fraudulent listings. For example, the Reuse Vandy group chat is notorious for having “sellers” that sell cheap concert tickets under the disguise of other Vanderbilt students. These are some of the limitations that were consistently reflected in our interviews and observational studies as you can see in Project1, Part 1.

Problem Undertaken (From P1 Needfinding):

The major problem identified from P1 is the lack of an efficient, searchable, safe and trustworthy virtual marketplace for Vanderbilt students.
Some of the key problems identified were:

- Severe clutter and unstructured posts; Inefficient search and filtering; Uncertainty about item availability; Repetitive buyer–seller communication; Lack of secure verification; Unsafe or unreliable meetups; Unclear or inconvenient payment processes; Limited accessibility and inconsistent browsing experience; Risk of scammers/bots; Limited trust between buyers and sellers; Missing or inconsistent information.

These problems formed the foundations for our design goals which directly formed our solution.

Design Goals (From P1 Needfinding):

- Fast, frictionless item discovery; Structured, complete listings; Clear availability status; Verified Vanderbilt-only accounts; Safer and more reliable hand-offs; Effective payment methods; Smart notifications and saved searches; Quick and accessible browsing.

Description of Final Solution:
Our final solution is a mobile application that Vanderbilt University can use to buy and sell used items within campus premises. With this mobile app, every student can upload items they want to sell with the details of the item (e.g. location for pickup, price of item e.t.c). Every buyer can also search items that they are interested in buying or scroll their feed to see available items and details of the seller. Further details of our solution would be specified below.
Our final solution for the ReDuce, ReUse, ReCycle mobile application directly solves each problem and design goals listed above through an intentional user interface and interactive design. Below is the high-description of our core features with appropriate screenshots of how our solution addressed the problems and design goals.

How Our Solution Solved Each Problem and Design Goal:

1. ***Verified and Secure Community:***

- Design Goals: Vanderbilt-only authentication
- Problems Solved: Lack of secure verification; Risk of scammers/bots; Limited trust between buyers and sellers.
- **Solution Feature:** Vanderbilt SSO login, in-app user reporting tools, and detailed buyer/seller profiles with ratings and recent activity to ensure all interactions occur within a verified, trustworthy community.

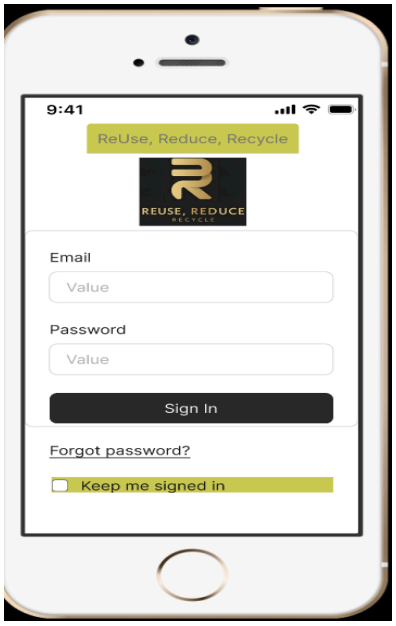


Figure 1

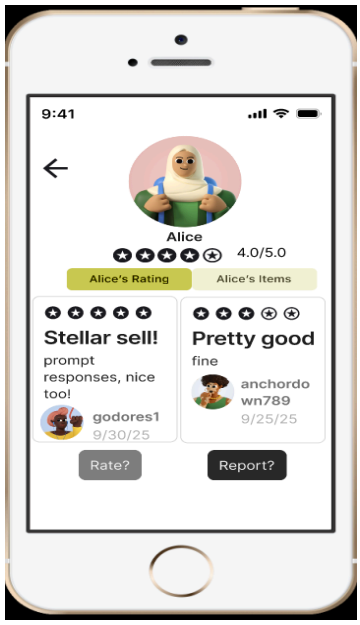


Figure 8

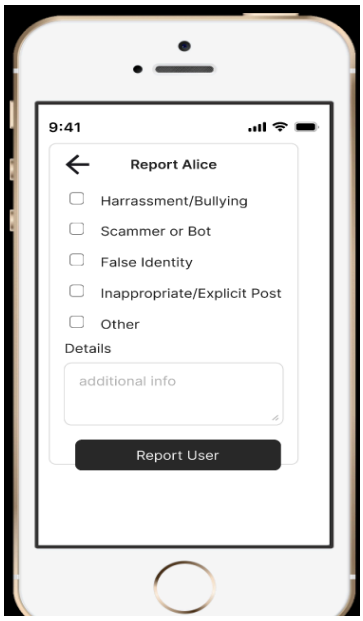


Figure 9

Figures 1, 8, and 9 show how our mobile app builds a secure, Vanderbilt only market place by using a mandatory SSO authentication which can only be used by Vanderbilt University student unique VUNET ID. This prevents bots and non-students from accessing the platform. As seen in figure 9, the report feature allows users flag inappropriate behaviors and from feedback from other class members, this feature also allows users to flag verified users that violate

guidelines. Each buyer and seller also have their profile pages that display ratings and user history, increasing transparency and user confidence. Overall, these features ensure safer and more accountable transactions for our users.

2. ***Efficient Item Discovery and Availability Transparency:***

- Design Goals: Fast, frictionless item discovery; Quick and accessible browsing; Clear availability status
- Problems Solved: Severe clutter and unstructured posts; Inefficient search and filtering; Limited accessibility and inconsistent browsing experience; Uncertainty about item availability.
- Solution Feature: A structured home feed with item cards, a built-in search bar, and quick-access navigation tabs that allow users to browse listings, search for specific items, and view seller and pickup details without scrolling through cluttered chats.
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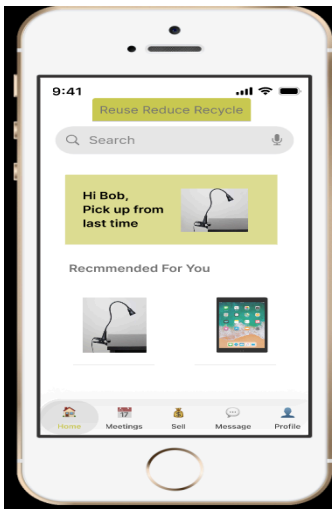


Figure 2

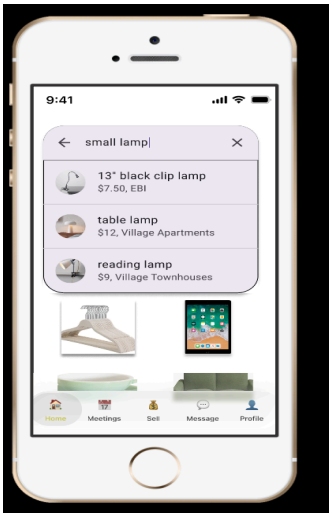


Figure 3

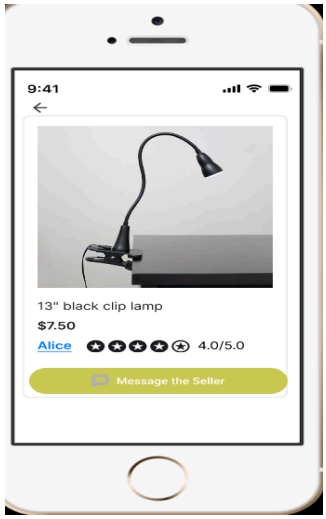


Figure 4

The screenshots above demonstrated how our redesigned interface enables fast frictionless item discovery by replacing the unstructured GroupMe chat with a clean, organised home feed that immediately displays relevant item cards containing photos, prices, seller information, and pickup locations. The built-in search bar allows users to quickly locate specific items e.g. “small lamp” without scrolling through cluttered messages, while the bottom navigation icons let users switch between browsing, selling, messaging and scheduling with ease. Once an item is sold, it automatically disappears from the search results and home feed which prevents students from wasting time on unavailable items. Together, these features directly address the inefficiencies identified in needfinding by eliminating excessive scrolling, reducing cognitive load and allowing students to find what they need quickly and reliably.

3. ***Streamlined Buyer–Seller Communication***

- Design Goals**:** Safer and more reliable hand-offs (coordination); Effective payment methods (coordination); Structured listings (supports messaging clarity).
- Problems Solved: Repetitive buyer–seller communication; Unsafe or unreliable meetups; Unclear or inconvenient payment processes
- Solution Feature: Integrated messaging that allows buyers to directly contact sellers from each listing to negotiate prices, clarify details, and coordinate logistics in one place.



Figure 5

- Figures 4 and 5 illustrate how RRR simplifies buyer–seller communication by allowing users to enter a conversation directly from the item page and coordinate all relevant details in one place. Because RRR does not include its own payment system, the messaging interface also provides a reliable channel for buyers and sellers to discuss preferred payment methods (e.g. Venmo, Zelle or in-person cash) without needing to switch to external apps. This unified communication flow reduces the scattered back and forth that currently occurs in other marketplace platforms, supports faster responses, and ensures all-transaction information is discussed clearly and efficiently within the same thread.

4. Safe Meetups

- Design Goals: Safer and more reliable hand-offs.
- Problems Solved: Unsafe meetups; Miscommunication about pickup times and locations.
- Solution Feature: A built-in scheduling system with a calendar view of all upcoming meetings, including time, location, and the user involved.



Figure 6

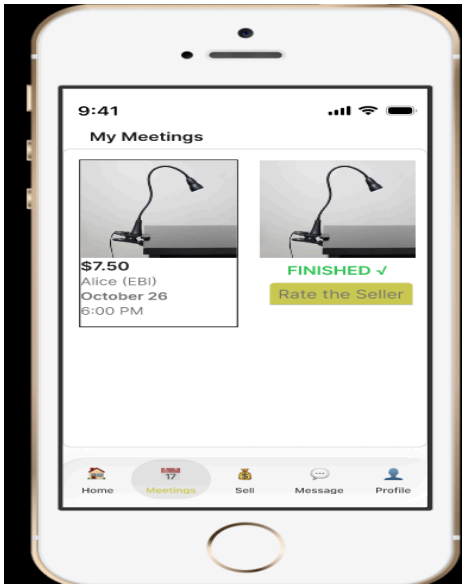


Figure 7

- Figure 6 and 7 shows how RRR supports safe and organized meetups by giving users a central calendar for all upcoming transactions and a summary of each communication (e.g. date of pickup, pickup location e.t.c.) in their own pages. Each scheduled meetup includes the location, time, and counterpart's name, reducing miscommunication and preventing no-shows. This scheduling structure creates safer, more reliable hand-offs compared to informal chat arrangements.

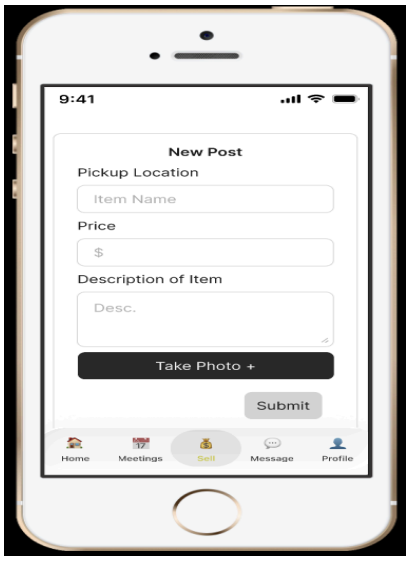
5. Smart Notification and Personalized Feeds

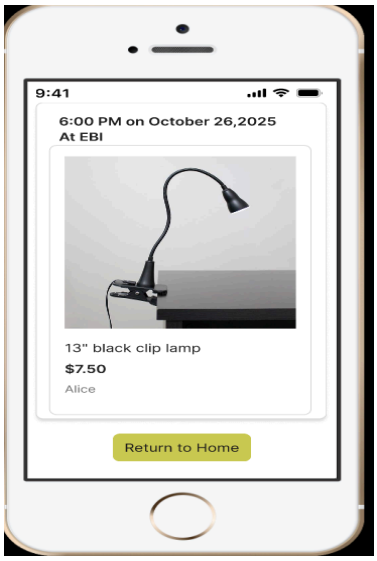
- Design Goals: Smart notifications and saved searches; Quick and accessible browsing
- Problems Solved: Missing deals due to message volume; Inefficient searching; Need to monitor frequently.
- Solution Feature: Personalized home feed recommendations, saved searches, and notifications for new items matching user interests.
- As you can see in Figure 2, the RRR app personalizes each user's home feed based on past searches and interactions. This ensures that relevant items appear immediately without requiring repeated searches. Users not only get recommendations but notifications on items they may be interested in based on recent activity.

6. Structured Listing

- Design Goals: Structured, complete listings
- Problems Solved: Missing or inconsistent information; Severe clutter and unstructured posts; Repetitive questions

- Solution Feature: A mandatory listing template requiring photos, price, pickup location, and other details related to the item to ensure every buyer receives the same essential information before messaging.





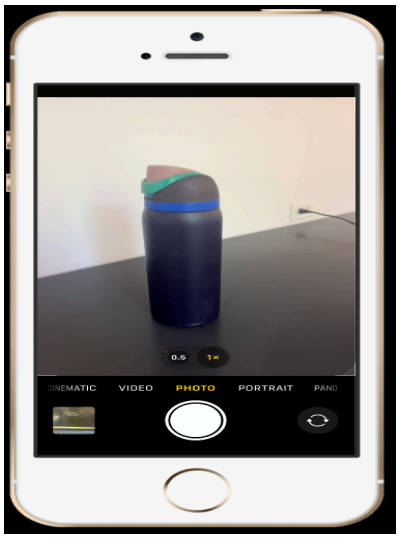


Figure 10 **Figure 11** **Figure 12 (Additional Image)**

- Figure 10 and 11 demonstrates how RRR standardizes the presentation of item details by requiring sellers to include all core information in each listing. This prevents the repetitive questions and clarification messages common in GroupMe, where posts often lack photos or essential detail

Overall Flow / Path:

<u>Buying an item:</u>

Figure 1 -> 2 -> 3 -> 4 -> 5 / 8 -> 6 -> 7 -> 11

<u>Selling an item:</u>

Figure 1 -> 10 -> 12 -> 13

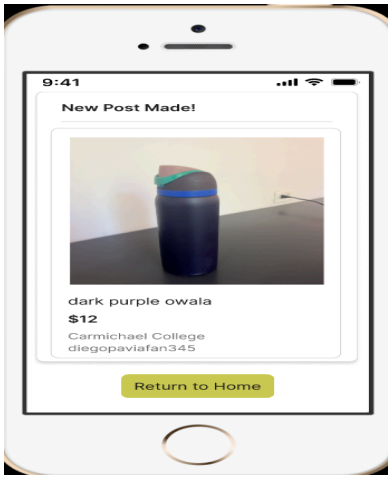


Figure 13

B. Design Solution Evolution:

1. Design Goal: Verified campus-only accounts

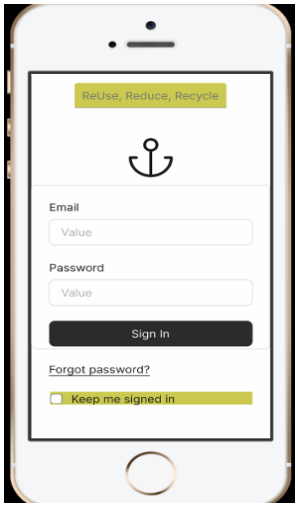
- Problem We Were Solving: Lack of secure verification; Risk of scammers/bots; Limited trust between buyers and sellers; Unreliable or unsafe transactions

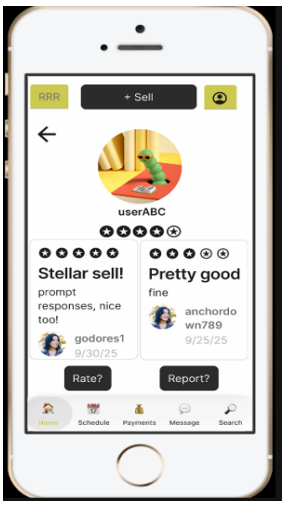
Low-Fidelity Sketch (Initial Concept): We began with a simple VUnetID login screen to restrict access to Vanderbilt students, but this alone did not address scammers who could still misuse the platform.

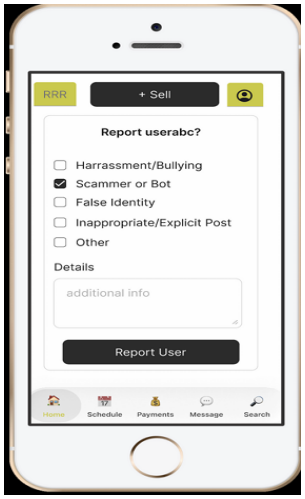


Figure I

Medium Fidelity Sketch (Strengthening Trust & Safety): We added screens for reporting users and rating buyers/sellers to strengthen trust and allow the community to self-moderate, reducing the risk of scam behavior.







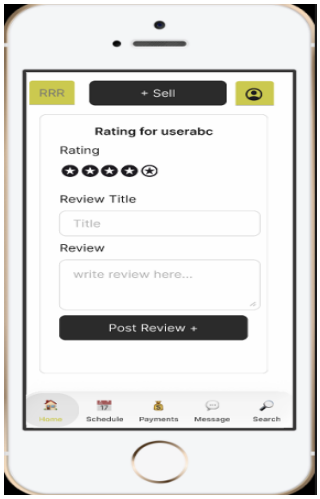


Figure II **Figure III** **Figure IV** **Figure V**

High Fidelity Sketch (Final Security Layer): We refined these features by adding a disabled/grey rating button for users who have not purchased from a seller and displaying the seller's active/past listings to prevent fake ratings and increase transparency.

As seen in Figure II, Figure III, Figure IV, Figure V and images below:



Figure VI

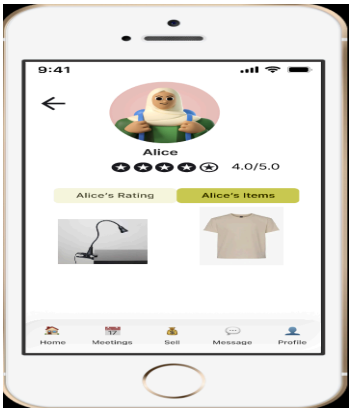


Figure VII

2. <u>Design Goal: Clear Availability Status of Items</u>

- <u>Problem We Were Solving:</u> Severe clutter and unstructured posts; Uncertainty about item availability.

- **Low-Fidelity Sketch (Initial Concept)**

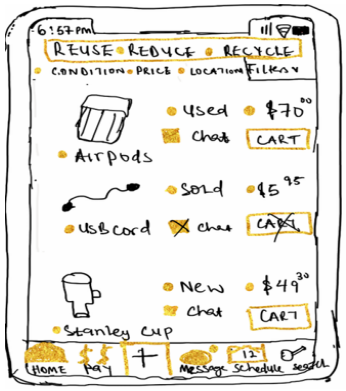


Figure I

The screen was crowded with item information, the search bar was placed too low, and sold items still appeared on the home page, making it difficult for users to identify available products.

- **Medium Fidelity Design**: We reorganized item cards, improved information hierarchy, moved the search bar to the top. However, the design was still not complete.



Figure II & III

High Fidelity Design: Sold Items were automatically removed from feed; we organized item cards, added our logo and brought down the sell button to reduce clutter on page.

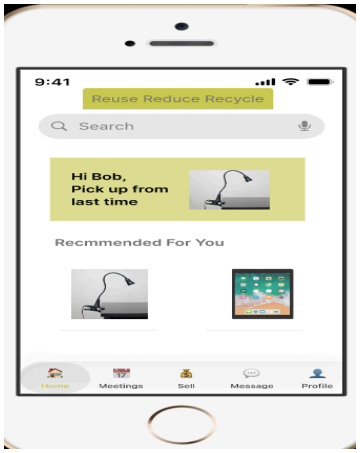


Figure IV



Figure V

C. Success Picture:

When evaluated against the design goals established in our Needfinding document, our final solution successfully meets the majority of the criteria we set at the beginning of the project. Each feature was intentionally designed to address specific problems identified through observations, interviews, cognitive walkthroughs, and peer reviews. As seen in the final interface, we achieved fast and frictionless item discovery, structured and complete listings, verified Vanderbilt-only access, safer hand-offs through integrated scheduling, and a reliable communication channel for buyers and sellers. These features directly resolved the pain points we encountered during research, such as cluttered information, uncertainty about item availability, safety concerns, and the prevalence of unverified interactions in the existing GroupMe system.

While our design meets most criteria, a few areas could be improved to fully align with our original goals. For example, although sold items are automatically removed from the feed, items that are “pending” (i.e., in the middle of negotiation or awaiting confirmation) are not flagged on the main listing.

Adding a visible “Pending” status on item cards or detail pages would provide fuller transparency for buyers. Additionally, while our system includes smart notifications and a personalized feed, it does not yet offer a dedicated interface for manually saving searches. Instead, users see a recent search history embedded in the search bar, which is functional but not fully aligned with the intended “saved searches” design goal.

Despite these limitations, the final design strongly reflects the principles emphasized throughout the course. Our team consistently grounded decisions in user needs, iterated based on feedback, and ensured that every major feature directly solved a documented problem. Overall, the project meets its core objectives and delivers a

significantly improved, research-driven alternative to the existing campus marketplace experience.

D. What We Did Well:

1. Challenge Faced: Difficulty Identifying a Pickup Time Both Parties Could Commit To

One major challenge that emerged during needfinding was the difficulty buyers and sellers faced when trying to agree on a pickup time and location. Users often discussed availability informally through messages, but this frequently resulted in no-shows because students forgot the time they agreed on or miscommunicated details. The challenge for our team was translating this recurring user pain point into a concrete design solution that reduced coordination friction and improved reliability.

To address this, we reviewed user feedback from interviews, where students suggested features such as shared calendars, reminders, and clearer scheduling tools. Based on these insights, we designed a built-in scheduling system (Figure 6) that allows buyers and sellers to select a mutually available time directly within the app. We also enabled users to add agreed-upon times from the messaging tab to their personal calendar with one tap, ensuring they receive reminders before the meetup. Additionally, we created an “Upcoming Transactions” page (Figure 7) that consolidates all scheduled pickups in one place, helping users keep track of their commitments and reducing forgotten meetings. This solution directly addressed the challenge, improved reliability, and supported our design goal of creating safer and more predictable hand-offs.

2. Challenge Faced: Balancing Information Density with Visual Clarity

   One challenge we encountered throughout the design process was finding the right balance between displaying enough information for users to make informed decisions and preventing the interface from becoming visually overwhelming. Our initial home screen included all item details on a single page, which made the interface crowded and difficult to scan. After multiple discussions, peer critiques, and reviewing HCI principles on information hierarchy, we decided to restructure the experience by minimizing text on the home feed and shifting detailed information to dedicated item pages that appear only when a user selects an item. This allowed us to simplify the visual layout, use clearer icons for navigation, and distribute information across multiple screens in a way that supported both clarity and usability.

E. What We Did Poorly:

Mistake 1: Underestimating Complexity of Item Status

A significant mistake we made during the design process was overlooking the full complexity of item status behavior, specifically the need for a “Pending” state. For most of the semester, we assumed items fell cleanly into only two categories: “Available” or “Sold.” However, during peer evaluations late in the high-fidelity phase, we realized that the negotiation period (i.e. when a seller is in conversation with a potential buyer but has not finalized the transaction) is a critical state that prevents other users from wasting time messaging about an item that is effectively spoken for.

Because we did not incorporate this “Pending” state during our low-fidelity sketching or early mid-fidelity iterations, we lacked UI elements to display it.This not only added friction for sellers but also created ambiguity for buyers when the “middle” state should have been clear, directly contradicting our design goal of improving transparency. By acting as if the item state was binary, we created confusion for our sellers/buyers.

Had we identified this need earlier, we could have integrated a proper “Pending” indicator into item cards and detail pages, ensuring a more accurate, reliable browsing experience. Earlier mapping of system states or writing a simple state diagram would likely have prevented this oversight.

Mistake 2: Overcrowded Early Interface Designs

In the initial sketching and low-fidelity phases, several of our screens (especially the home feed and item detail pages) contained far too much cluttered information. We attempted to include all item attributes directly on the feed rather than making use of progressive disclosure. This resulted in disorganized layouts that violated basic HCI principles around readability, scanning, and cognitive load.

While we eventually recognized the issue and redesigned the views to separate high-level and detailed information, the process took longer because we had not validated the complexity of our early designs with quick user walkthroughs. Faster feedback loops with even informal user tests could have helped us detect the overload sooner and prevented multiple redesign cycles. Looking back, we would approach our early interface designs with much more simplicity and just have separate pages for different use cases/user tasks.

F. Other Lessons

Throughout the design of RRR, several broader HCI and design lessons became clear. These are lessons that extend far beyond this project and will shape how we approach future UI/UX design work.

One of the clearest lessons was how foundational early design decisions are. Overlooking the “Pending” item state early on created significant downstream challenges, forcing us to retrofit logic into an already-polished interface. This

demonstrated that even seemingly small modeling choices (e.g., item states, role-based flows, navigation hierarchy) can dramatically shape the usability and maintainability of the final product. In future projects, we would invest more time in early systems thinking (e.g. mapping states, transitions, and edge cases before beginning visual design).

Our experience with the seller flow reinforced that intuition is not an adequate substitute for user testing. As designers, we assumed the listing flow was “simple enough,” but actual users quickly revealed friction points we hadn’t anticipated. This highlighted a core HCI principle: designers grow blind to complexity as they become familiar with a system. Real users, especially first-time users, expose misunderstandings, confusion, and gaps that designers cannot predict on their own. Moving forward, we would prioritize lightweight, continuous user testing from the earliest sketches to avoid late-stage redesigns.

We learned that iteration is not simply about making the design “look better” but about strategically refining workflows to match user needs (despite design being so much about aesthetic appeal, it’s ultimately about usability). Many of our strongest improvements (such as reorganizing item cards, clarifying information hierarchy, or separating item details across multiple screens) only emerged after multiple cycles of critique and redesign. This emphasized the importance of being willing to discard or drastically revise earlier ideas, even ones we initially felt confident about.

Throughout the project, we encountered recurring tension between visual clarity and information density. Our early screens leaned too heavily toward completeness, while later iterations risked oversimplifying. We learned that the best designs strike a balance between aesthetics and function, surfacing key information without overwhelming users, and hiding details only when they are not immediately relevant. Achieving this balance requires both careful planning and repeated refinement.

A final lesson we learned is the value of consistent collaboration throughout a complex design project. Our team worked well together, and our shared discussions, critiques, and brainstorming sessions often generated insights that none of us would have reached alone. Even when we had different perspectives on features or visual direction, these differences ultimately strengthened the final product by pushing us to justify decisions and think more deeply about user needs. This project reinforced that good design is a collective effort that benefits from diverse viewpoints, open communication, and a willingness to iterate together. In future projects, we would continue this collaborative approach, pairing it with earlier documentation and shared planning to make our workflow even more efficient. For a project about how well “others” can navigate and use an app, it was incredibly helpful to have a group of four people rather than have this project be an individual effort.